

# Chapter 5, Section 2

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1. If  $X$  is a birationally ruled surface, show that the curve  $C$ , such that  $X$  is birationally equivalent to  $C \times \mathbf{P}^1$ , is unique up to isomorphism.
2. Let  $X$  be the ruled surface  $\mathbf{P}(\mathcal{E})$  over a curve  $C$ . Show that  $\mathcal{E}$  is decomposable if and only if there exist two sections  $C', C''$  of  $X$  such that  $C' \cap C'' = \emptyset$ .
3. (a) If  $\mathcal{E}$  is a locally free sheaf of rank  $r$  on a nonsingular curve  $C$ , then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that  $\mathcal{E}_i/\mathcal{E}_{i-1}$  is an invertible sheaf for each  $i = 1, \dots, r$ . We say that  $\mathcal{E}$  is a *successive extension* of invertible sheaves.

- (b) Show that this is false for varieties of dimension  $\geq 2$ . In particular, the sheaf of differentials  $\Omega_{\mathbf{P}^2}$  on  $\mathbf{P}^2$  is not an extension of invertible sheaves.
4. Let  $C$  be a curve of genus  $g$ , and let  $X$  be the ruled surface  $C \times \mathbf{P}^1$ . We consider the question, for what integers  $s \in \mathbf{Z}$  does there exist a section  $D$  of  $X$  with  $D^2 = s$ ? First show that  $s$  is always an even integer, say  $s = 2r$ .
  - (a) Show that  $r = 0$  and  $r \geq g + 1$  are always possible.
  - (b) If  $g = 3$ , show that  $r = 1$  is not possible, and just one of the two values  $r = 2, 3$  is possible, depending on whether  $C$  is hyperelliptic or not.
5. *Values of  $e$ .* Let  $C$  be a curve of genus  $g \geq 1$ .

- (a) Show that for each  $0 \leq e \leq 2g - 2$  there is a ruled surface  $X$  over  $C$  with invariant  $e$ , corresponding to an indecomposable  $\mathcal{E}$ .
- (b) Let  $e < 0$ , let  $D$  be any divisor of degree  $d = -e$ , and let  $\xi \in H^1(X, \mathcal{O}_X(-D))$  be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_X(D) \longrightarrow 0.$$

Let  $H \subseteq |D + K|$  be the sublinear system of codimension 1 defined by  $\ker \xi$ , where  $\xi$  is considered as a linear functional on  $H^0(X, \mathcal{O}_X(D + K))$ . For any effective divisor  $E$  of degree  $d - 1$ , let  $L_E \subseteq |D + K|$  be the sublinear system  $|D + K - E| + E$ . Show that  $\mathcal{E}$  is normalized if and only if for each  $E$  as above,  $L_E \not\subseteq H$ .

- (c) Now show that if  $-g \leq e < 0$ , there exists a ruled surface  $X$  over  $C$  with invariant  $e$ .  
 (d) For  $g = 2$ , show that  $e \geq -2$  is also necessary for the existence of  $X$ .

6. Show that every locally free sheaf of finite rank on  $\mathbf{P}^1$  is isomorphic to a direct sum of invertible sheaves.  
 7. On the elliptic ruled surface  $X$  of (2.11.6), show that the sections  $C_0$  with  $C_0^2 = 1$  form a one-dimensional algebraic family, parametrized by the points of the base curve  $C$ , and that no two are linearly equivalent.  
 8. A locally free sheaf  $\mathcal{E}$  on a curve  $C$  is said to be *stable* if for every quotient locally free sheaf  $\mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$ ,  $\mathcal{F} \neq \mathcal{E}$ ,  $\mathcal{F} \neq 0$ , we have

$$(\deg \mathcal{F})/\text{rank } \mathcal{F} > (\deg \mathcal{E})/\text{rank } \mathcal{E}$$

Replacing  $>$  by  $\geq$  defines *semistable*.

- (a) A decomposable  $\mathcal{E}$  is never stable.  
 (b) If  $\mathcal{E}$  has rank 2 and is normalized, then  $\mathcal{E}$  is stable (respectively, semistable) if and only if  $\deg \mathcal{E} \leq 0$  (respectively,  $\geq 0$ ).  
 (c) Show that the indecomposable locally free sheaves  $\mathcal{E}$  of rank 2 that are not semistable are classified, up to isomorphism, by giving (1) an integer  $0 < e \leq 2g - 2$ , (2) an element  $\mathcal{L} \in \text{Pic } C$  of degree  $-e$ , and (3) a nonzero  $\xi \in H^1(X, \mathcal{L}^{-1})$ , determined up to a nonzero scalar multiple.
9. Let  $Y$  be a nonsingular curve on a quadric cone  $X_0$  in  $\mathbf{P}^3$ . Show that either  $Y$  is a complete intersection of  $X_0$  with a surface of degree  $a \geq 1$ , in which case  $\deg Y = 2a$ ,  $g(Y) = (a - 1)^2$ , or  $\deg Y$  is odd, say  $2a + 1$ , and  $g(Y) = a^2 - a$ .
10. For any  $n > e \geq 0$ , let  $X$  be the rational scroll of degree  $d = 2n - e$  in  $\mathbf{P}^{d+1}$  given by (2.19). If  $n \geq 2e - 2$ , show that  $X$  contains a nonsingular curve  $Y$  of genus  $g = d + 2$  which is a canonical curve in this embedding. Conclude that for every  $g \geq 4$ , there exists a nonhyperelliptic curve which has a  $g_3^1$ .
11. Let  $X$  be a ruled surface over the curve  $C$ , defined by a normalized bundle  $\mathcal{E}$ , and let  $\mathfrak{e}$  be the divisor on  $C$  for which  $\mathcal{O}_C(\mathfrak{e}) \cong \wedge^2 \mathcal{E}$  (2.8.1). Let  $\mathfrak{b}$  be any divisor on  $C$ .
- (a) If  $|\mathfrak{b}|$  and  $|\mathfrak{b} + \mathfrak{e}|$  have no base points, and if  $\mathfrak{b}$  is nonspecial, then there is a section  $D \sim C_0 + \mathfrak{b}f$ , and  $|D|$  has no points.  
 (b) If  $\mathfrak{b}$  and  $\mathfrak{b} + \mathfrak{e}$  are very ample on  $C$ , and for every point  $P \in C$ , we have  $\mathfrak{b} - P$  and  $\mathfrak{b} + \mathfrak{e} - P$  nonspecial, then  $C_0 + \mathfrak{b}f$  is very ample.
12. Let  $X$  be a ruled surface with invariant  $e$  over an elliptic curve  $C$ , and let  $\mathfrak{b}$  be a divisor on  $C$ .
- (a) If  $\deg \mathfrak{b} \geq \mathfrak{e} + 2$ , then there is a section  $D \sim C_0 + \mathfrak{b}f$  such that  $|D|$  has no base points.  
 (b) The linear system  $|C_0 + \mathfrak{b}f|$  is very ample if and only if  $\deg \mathfrak{b} \geq e + 3$ .

13. For every  $e \geq -1$  and  $n \geq e + 3$ , there is an elliptic scroll of degree  $d = 2n - e$  in  $\mathbf{P}^{d_1}$ . In particular, there is an elliptic scroll of degree 5 in  $\mathbf{P}^4$ .
14. Let  $X$  be a ruled surface over a curve of genus  $g$ , with invariant  $e < 0$ , and assume that  $\text{char } k = p > 0$  and  $g \geq 2$ .
- (a) If  $Y \equiv aC_0 + bf$  is an irreducible curve  $\neq C_0 \cdot f$ , then either  $a = 1, b \geq 0$ , or  $2 \leq a \leq p-1, b \geq \frac{1}{2}ae$ , or  $a \geq p, b \geq \frac{1}{2}ae + 1 - g$ .
  - (b) If  $a > 0$  and  $b > a(\frac{1}{2}e + (1/p)(g-1))$ , then any divisor  $D \equiv aC_0 + bf$  is ample. On the other hand, if  $D$  is ample, then  $a > 0$  and  $b > \frac{1}{2}ae$ .
15. *Funny behaviour in characteristic  $p$ .* Let  $C$  be the plane curve  $x^3y + y^3z + z^3x = 0$  over a field  $k$  of characteristic 3.
- (a) Show that the action of the  $k$ -linear Frobenius morphism  $f$  on  $H^1(C, \mathcal{O}_C)$  is identically 0.
  - (b) Fix a point  $P \in C$ , and show that there is a nonzero  $\xi \in H^1(C, \mathcal{O}_C(-P))$  such that  $f^*\xi = 0$  in  $H^1(C, \mathcal{O}_C(-3P))$ .
  - (c) Now let  $\mathcal{E}$  be defined by  $\xi$  as an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0,$$

and let  $X$  be the corresponding ruled surface over  $C$ . Show that  $X$  contains a nonsingular curve  $Y \equiv 3C_0 - 3f$ , such that  $\pi : Y \rightarrow C$  is purely inseparable. Show that the divisor  $D = 2C_0$  satisfies the hypothesis of (2.21b), but is not ample.

16. Let  $C$  be a nonsingular affine curve. Show that two locally free sheaves  $\mathcal{E}, \mathcal{E}'$  of the same rank are isomorphic if and only if their classes in the Grothendieck group  $K(X)$  are the same. This is false for a projective curve.
17. (a) Let  $\varphi : \mathbf{P}_k^1 \rightarrow \mathbf{P}_k^3$  be the 3-uple embedding. Let  $\mathcal{I}$  be the sheaf of ideals of the twisted cubic curve  $C$  which is the image of  $\varphi$ . Then  $\mathcal{I}/\mathcal{I}^2$  is a locally free sheaf of rank 2 on  $C$ , so  $\varphi^*(\mathcal{I}/\mathcal{I}^2)$  is a locally free sheaf of rank 2 on  $\mathbf{P}^1$ . By (2.14), therefore,  $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$  for some  $l, m \in \mathbf{Z}$ . Determine  $l$  and  $m$ .
- (b) Repeat part (a) for the embedding  $\varphi : \mathbf{P}^1 \rightarrow \mathbf{P}^3$  given by  $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$ , whose image is a nonsingular rational quartic curve.